EXTENSIVE MATHEMATICAL DERIVATIONS

Brownian Bridge Diffusion Model

Forward Process and Markov Transition

Step-by-step proofs, probability identities,
Gaussian calculations, and mathematical interpretation

PART 1 — FOUNDATIONS AND MARKOV TRANSITION

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1 Scope and notation

This report establishes the probabilistic foundations of the Brownian Bridge Diffusion Model (BBDM), beginning with its endpoint-conditioned forward marginal and deriving the Markov transition between consecutive timesteps. Particular care is taken to distinguish marginal Gaussian noises from independent innovations. This distinction is essential: the marginal equations alone do not imply independent noise at different timesteps.

We use $x_0 \in \mathbb{R}^d$ for the source data, $y \in \mathbb{R}^d$ for the conditioning endpoint, and x_t for the state at timestep t. Define

$$a_t = 1 - m_t.$$

The BBDM forward marginal is

$$x_t = a_t x_0 + m_t y + B_t, \qquad B_t \sim \mathcal{N}(0, \delta_t I), \tag{1.1}$$

or, equivalently,

$$x_t = (1 - m_t)x_0 + m_t y + \sqrt{\delta_t}\,\epsilon_t, \qquad \epsilon_t \sim \mathcal{N}(0, I). \tag{1.2}$$

The scalar δ_t is the marginal noise variance; $\sqrt{\delta_t}$ is its standard-deviation multiplier.

1.1 Marginal distributions

Conditioning on fixed endpoints x_0, y, the forward marginal is

$$q_{BB}(x_t \mid x_0, y) = \mathcal{N}(x_t;\, a_t x_0 + m_t y,\, \delta_t I)\,. \tag{1.3}$$

At the preceding time,

$$q_{BB}(x_{t-1} \mid x_0, y) = \mathcal{N}(x_{t-1};\, a_{t-1} x_0 + m_{t-1} y,\, \delta_{t-1} I)\,. \tag{1.4}$$

Both are Gaussian marginals. However, knowing the two marginal distributions does not by itself determine their joint distribution: a cross-covariance is also required.

2 Why consecutive marginal noises are correlated

A frequent but incorrect shortcut is to treat ϵ_t and ϵ_{t-1} in (1.2) as independent. They are standard Gaussian marginal variables, but in a Brownian Bridge construction they are generally correlated. The Markov process is more naturally written using a new independent innovation at every transition.

Center the process at the target endpoint:

$$B_t = x_t - (a_t x_0 + m_t y).$$

Then

$$B_t \sim \mathcal{N}(0, \delta_t I). \tag{2.1}$$

For the Brownian Bridge Markov coupling, the conditional mean has the bridge contraction form

$$\mathbb{E}[B_t \mid B_{t-1}] = r_t B_{t-1}, \qquad r_t = \frac{a_t}{a_{t-1}} = \frac{1-m_t}{1-m_{t-1}}. \tag{2.2}$$

The ratio r_t is the ratio of the remaining deterministic displacement from the endpoint at the two timesteps:

$$\frac{a_t(x_0 - y)}{a_{t-1}(x_0 - y)} = \frac{a_t}{a_{t-1}}.$$

This explains the interpolation ratio. The stochastic conditional-mean relation in (2.2), however, additionally uses the Brownian Bridge Markov covariance structure; it does not follow from deterministic interpolation alone.

2.1 Deriving the cross-covariance

For square-integrable random vectors X, Y, the law of total covariance gives

$$\mathrm{Cov}(X, Y) = \mathbb{E}[\mathrm{Cov}(X, Y \mid Y)] + \mathrm{Cov}(\mathbb{E}[X \mid Y], \mathbb{E}[Y \mid Y]).$$

Since Y is known when conditioning on itself,

$$\mathrm{Cov}(X, Y \mid Y) = 0, \qquad \mathbb{E}[Y \mid Y] = Y.$$

Consequently,

$$\mathrm{Cov}(X, Y) = \mathrm{Cov}(\mathbb{E}[X \mid Y], Y). \tag{2.3}$$

Apply this with $X = B_t$ and $Y = B_{t-1}$:

$$\begin{aligned} \mathrm{Cov}(B_t, B_{t-1}) &= \mathrm{Cov}(\mathbb{E}[B_t \mid B_{t-1}], B_{t-1}) \\ &= \mathrm{Cov}(r_t B_{t-1}, B_{t-1}) \\ &= r_t \,\mathrm{Cov}(B_{t-1}, B_{t-1}) \\ &= r_t \delta_{t-1} I. \end{aligned} \tag{2.4}$$

Since $B_t = \sqrt{\delta_t}\epsilon_t$ and $B_{t-1} = \sqrt{\delta_{t-1}}\epsilon_{t-1}$,

$$\boxed{\mathrm{Cov}(\epsilon_t, \epsilon_{t-1}) = r_t \sqrt{\frac{\delta_{t-1}}{\delta_t}}\, I = \frac{1-m_t}{1-m_{t-1}} \sqrt{\frac{\delta_{t-1}}{\delta_t}}\, I.} \tag{2.5}$$

Thus the marginal noises are correlated, even though each has a standard-normal marginal distribution.

3 The Markov transition and the origin of the minus sign

The conditional Gaussian distribution of B_t given B_{t-1} has mean $r_t B_{t-1}$. Its covariance is obtained from the conditional covariance formula for a jointly Gaussian pair:

$$\mathrm{Var}(B_t \mid B_{t-1}) = \mathrm{Var}(B_t) - \mathrm{Cov}(B_t, B_{t-1})\,\mathrm{Var}(B_{t-1})^{-1}\,\mathrm{Cov}(B_{t-1}, B_t).$$

Substituting

$$\mathrm{Var}(B_t) = \delta_t I, \quad \mathrm{Var}(B_{t-1}) = \delta_{t-1} I, \quad \mathrm{Cov}(B_t, B_{t-1}) = r_t \delta_{t-1} I,$$

we get

$$\begin{aligned}\mathrm{Var}(B_t \mid B_{t-1}) &= \delta_t I - (r_t \delta_{t-1} I)(\delta_{t-1}^{-1} I)(r_t \delta_{t-1} I) \\ &= (\delta_t - r_t^2 \delta_{t-1}) I.\end{aligned}$$

Define

$$\boxed{\delta_{t|t-1} = \delta_t - r_t^2 \delta_{t-1}.} \tag{3.1}$$

The minus sign is therefore a consequence of conditioning a correlated Gaussian pair. It is not obtained by adding two independent marginal noise variances.

Hence

$$B_t \mid B_{t-1} \sim \mathcal{N}\Big(r_t B_{t-1},\, \delta_{t|t-1} I\Big)\,. \tag{3.2}$$

Returning to the uncentered state,

$$\begin{aligned}x_t &= a_t x_0 + m_t y + B_t \\ &= a_t x_0 + m_t y + r_t(x_{t-1} - a_{t-1} x_0 - m_{t-1} y) + \sqrt{\delta_{t|t-1}}\, z_t \\ &= (a_t - r_t a_{t-1}) x_0 + (m_t - r_t m_{t-1}) y + r_t x_{t-1} + \sqrt{\delta_{t|t-1}}\, z_t.\end{aligned}$$

Because $r_t = a_t / a_{t-1}$, the x_0 coefficient vanishes:

$$a_t - r_t a_{t-1} = 0.$$

Define

$$b_t = m_t - r_t m_{t-1}.$$

The transition is therefore

$$\boxed{q_{BB}(x_t \mid x_{t-1}, y) = \mathcal{N}\Big(x_t;\, r_t x_{t-1} + b_t y,\, \delta_{t|t-1} I\Big)\,,} \tag{3.3}$$

where

$$r_t = \frac{1 - m_t}{1 - m_{t-1}}, \quad b_t = m_t - r_t m_{t-1}, \quad \delta_{t|t-1} = \delta_t - r_t^2 \delta_{t-1}. \tag{3.4}$$

Notice that x_0 no longer appears explicitly in the transition. Its contribution has been eliminated by expressing the next state in terms of the current state and the endpoint.

3.1 Independent innovations versus marginal noises

Equation (3.2) can be sampled as

$$B_t = r_t B_{t-1} + \sqrt{\delta_{t|t-1}}\, z_t, \qquad z_t \sim \mathcal{N}(0, I), \tag{3.5}$$

where z_t is independent of B_{t-1}. In terms of the marginal-noise variables,

$$\boxed{\sqrt{\delta_t}\epsilon_t = r_t \sqrt{\delta_{t-1}}\epsilon_{t-1} + \sqrt{\delta_{t|t-1}}\, z_t.} \tag{3.6}$$

The two terms on the right are independent. Taking variances,

$$\begin{aligned}\mathrm{Var}(\mathrm{RHS}) &= r_t^2\delta_{t-1}I + \delta_{t|t-1}I \\ &= r_t^2\delta_{t-1}I + (\delta_t - r_t^2\delta_{t-1})I \\ &= \delta_t I,\end{aligned}$$

which matches $\mathrm{Var}(\sqrt{\delta_t}\epsilon_t)$. This is the precise distinction:

- ϵ_t is the standardized marginal noise at time t;
- z_t is a fresh, independent innovation used to advance the Markov chain.

4 Specialization to the BBDM schedule

The schedule used in BBDM is

$$m_t = \frac{t}{T}, \qquad \delta_t = 2s(m_t - m_t^2) = 2s\, m_t(1 - m_t), \tag{4.1}$$

where $s > 0$ controls the variance scale. Then

$$r_t = \frac{1 - t/T}{1 - (t-1)/T} = \frac{T - t}{T - t + 1}. \tag{4.2}$$

For the transition variance,

$$\delta_{t|t-1} = \delta_t - r_t^2\delta_{t-1}.$$

Substituting the schedule and simplifying gives

$$\boxed{\delta_{t|t-1} = \frac{2s(T-t)}{T(T-t+1)}}. \tag{4.3}$$

An equivalent expression is

$$\delta_{t|t-1} = 2s\frac{(m_t - m_{t-1})(1 - m_t)}{1 - m_{t-1}}. \tag{4.4}$$

The transition mean coefficient on y simplifies to

$$b_t = m_t - r_t m_{t-1} = \frac{1}{T - t + 1}. \tag{4.5}$$

These expressions are valid for $1 \le t < T$; the terminal endpoint requires the usual limiting/boundary treatment because $1 - m_T = 0$.

5 Connection with the standard Brownian Bridge

For a standard Brownian Bridge on $[0, 1]$, let $m_t = t$ and $\delta_t = t(1 - t)$. For two times $s < t$,

$$r_{t|s} = \frac{1 - t}{1 - s}.$$

The conditional variance is

$$\begin{aligned}\delta_{t|s} &= t(1-t) - \left(\frac{1-t}{1-s}\right)^2 s(1-s) \\ &= (1-t)\left[t - \frac{s(1-t)}{1-s}\right] \\ &= (1-t)\frac{t(1-s)-s(1-t)}{1-s} \\ &= \frac{(t-s)(1-t)}{1-s}.\end{aligned}$$

Thus

$$\boxed{\delta_{t|s} = \frac{(t-s)(1-t)}{1-s}.} \tag{5.1}$$

This recovers the standard Brownian Bridge transition variance.

6 What the marginal equations do and do not establish

The two marginal statements

$$x_t \mid x_0, y \sim \mathcal{N}(a_t x_0 + m_t y, \delta_t I), \quad x_{t-1} \mid x_0, y \sim \mathcal{N}(a_{t-1} x_0 + m_{t-1} y, \delta_{t-1} I)$$

do not uniquely determine the joint law of (x_t, x_{t-1}). A joint Gaussian also requires the cross-covariance. The transition (3.3) follows when the Brownian Bridge Markov coupling is imposed, equivalently when the cross-covariance has the structure in (2.4).

The BBDM paper presents the forward marginals and the Markov transition formula, but the cross-time noise covariance derivation is not written out in the equations considered here. It is therefore best to label (2.4)–(2.5) as the mathematical implication of the Brownian Bridge Markov construction, rather than claiming that the paper explicitly states independent or correlated ϵ_t variables.